

# Project 3: Preference Agent - PDF Report

**Author:** Thomas Seife

**Date:** 3/27/2025

## 1 Introduction

This report presents the **Preference Agent (PrefAgent)** system developed for solving a preference problem using **binary attributes**, **hard constraints**, and **preference rules** in either **penalty logic** or **qualitative choice logic**. The system encodes all possible objects, checks feasibility using hard constraints, and then applies the chosen preference logic to compare and optimize these objects.

This project was developed in Python 3.11, using the PySAT library for satisfiability checking. Below, the system's design, implementation, and results obtained from a custom test instance are discussed.

## 2 System Workflow Overview

### 2.1 Input Files

- **Attributes File** (e.g., `TestCase/attributes.txt`): Lists all attributes and their two possible values.
- **Hard Constraints File** (e.g., `TestCase/constraints.txt`): Specifies the propositional constraints in CNF.
- **Preference File** (e.g., `TestCase/penaltylogic.txt` or `TestCase/qualitativechoicelogic.txt`): Specifies preferences either as penalty rules or qualitative choice rules.

### 2.2 Generating All Possible Objects

The system creates every possible combination of attribute values. Each combination is assigned a binary encoding (e.g., o0, o1, o2, ...).

### 2.3 Feasibility Checking

The system uses PySAT to test which objects are consistent with all hard constraints. Any infeasible objects are removed from further consideration.

## 2.4 Preference Evaluation

- **Penalty Logic:** Assigns penalty scores to feasible objects based on each rule and sums them up to compare overall scores.
- **Qualitative Choice Logic:** Compares objects pairwise based on “better-than” rules to determine if one object dominates another.

## 2.5 Output Tasks

The system performs several tasks:

1. **Encoding of Objects:** Displays all objects and their attributes.
2. **Feasibility Check:** Reports the number of objects that satisfy the hard constraints.
3. **Show the Table:** Presents a table of feasible objects with penalty or qualitative scores.
4. **Exemplification:** Randomly selects two feasible objects and compares their preference outcomes.
5. **Omni-optimization:** Identifies the optimal objects based on the chosen preference logic.

# 3 System Execution and Screenshots

## 3.1 Launching the Program

When the program is executed with:

```
python3.11 main.py
```

a welcome message is displayed with an ASCII art introduction (see Figure 1).

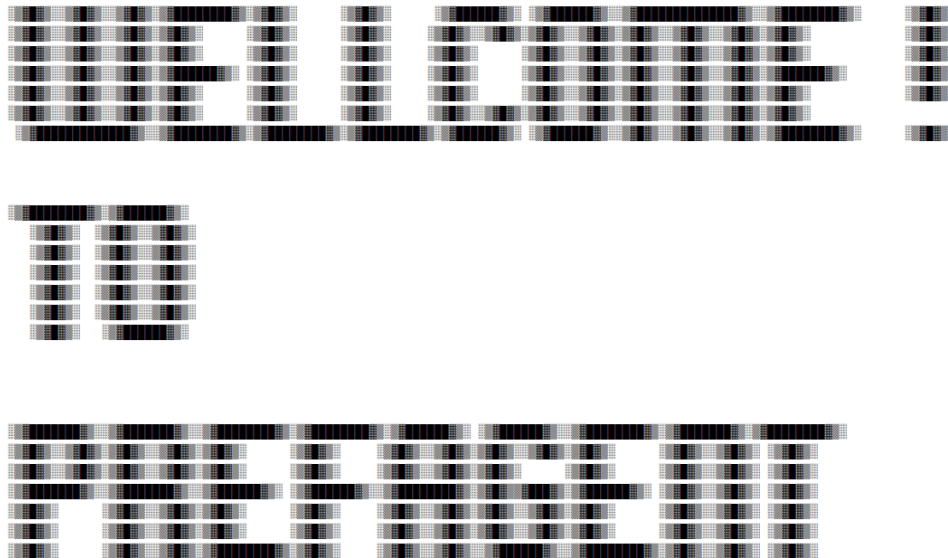


Figure 1: Welcome ASCII Art

### 3.2 Selecting Files and Preference Logic

The program prompts the user for:

- The **attributes file** (e.g., `TestCase/attributes.txt`)
- The **constraints file** (e.g., `TestCase/constraints.txt`)
- The type of **preference logic** (Penalty or Qualitative)

Figure 2 illustrates an example of the console prompts for file selection and preference logic.

```

Enter the attributes file name: TestCase/attributes.txt
Enter the Hard constraints file name: TestCase/constraints.txt
1. Penalty Logic
2. Qualitative Logic
3. Exit
Choose an option: 1
You picked Penalty Logic.
Enter the preference file name: TestCase/penaltylogic.txt

    Choose an option:
    1. Encoding
    2. Feasibility checking
    3. Show the Table
    4. Exemplification
    5. Omni-optimization
    6. Back to previous menu
Enter your choice: 1
Encoding of Objects:
o0 - ice-cream, beer, beef, chips
o1 - ice-cream, beer, beef, fries
o2 - ice-cream, beer, fish, chips
o3 - ice-cream, beer, fish, fries
o4 - ice-cream, wine, beef, chips
o5 - ice-cream, wine, beef, fries
o6 - ice-cream, wine, fish, chips
o7 - ice-cream, wine, fish, fries
o8 - cake, beer, beef, chips
o9 - cake, beer, beef, fries
o10 - cake, beer, fish, chips
o11 - cake, beer, fish, fries
o12 - cake, wine, beef, chips
o13 - cake, wine, beef, fries
o14 - cake, wine, fish, chips
o15 - cake, wine, fish, fries

```

Figure 2: Console Prompts for File Names and Preference Logic Selection

### 3.3 Penalty Logic Menu

After choosing **Penalty Logic** and providing a preference file (e.g., *TestCase/penaltylogic.txt*), the system displays a menu similar to the following:

1. Encoding
2. Feasibility checking
3. Show the Table

4. Exemplification
5. Omni-optimization
6. Back to previous menu

Figure 3 shows a sample console session with task selections and results.

```

Choose an option:
1. Encoding
2. Feasibility checking
3. Show the Table
4. Exemplification
5. Omni-optimization
6. Back to previous menu
Enter your choice: 2
Feasible Objects (9):

Choose an option:
1. Encoding
2. Feasibility checking
3. Show the Table
4. Exemplification
5. Omni-optimization
6. Back to previous menu
Enter your choice: 3
+-----+-----+-----+-----+-----+
| encoding | fish AND chips | wine OR ice-cream | NOT fries | total penalty |
+-----+-----+-----+-----+-----+
| o0       | 10             | 0                 | 0         | 10            |
| o1       | 10             | 0                 | 4         | 14            |
| o3       | 10             | 0                 | 4         | 14            |
| o4       | 10             | 0                 | 0         | 10            |
| o5       | 10             | 0                 | 4         | 14            |
| o7       | 10             | 0                 | 4         | 14            |
| o8       | 10             | 7                 | 0         | 17            |
| o9       | 10             | 7                 | 4         | 21            |
| o11      | 10             | 7                 | 4         | 21            |
+-----+-----+-----+-----+-----+

```

Figure 3: Penalty Logic Menu and Sample Outputs

### 3.3.1 Encoding (Choice 1)

The system lists all possible objects (e.g., o0, o1, o2, ...) along with their attribute values.

#### Example Output:

Encoding of Objects:

```

o0 -- ice-cream, beer, beef, chips
o1 -- ice-cream, beer, fish, fries
...

```

### 3.3.2 Feasibility Checking (Choice 2)

The system reports the number of objects that are feasible under the hard constraints.

#### Example Output:

Feasible Objects (9):

### 3.3.3 Show the Table (Choice 3)

A table is printed that displays each feasible object, the penalty contribution of each rule, and the total penalty.

| Encoding | Rule 1: FISH AND chips | Rule 2: wine OR ice-cream | Total Penalty |
|----------|------------------------|---------------------------|---------------|
| o0       | 10                     | 0                         | 14            |
| o1       | 0                      | 10                        | 14            |
| :        | :                      | :                         | :             |

Table 1: Sample Penalty Table

### 3.3.4 Exemplification (Choice 4)

The system randomly selects two feasible objects and displays the preference outcome (e.g., one object is strictly preferred over the other, they are equivalent, or they are incomparable).

### 3.3.5 Omni-optimization (Choice 5)

The system lists all optimal objects (i.e., objects with the minimum total penalty).

## 3.4 Qualitative Choice Logic Menu

When **Qualitative Choice Logic** is selected, the program prompts for the corresponding preference file (e.g., `TestCase/qualitativechoicelogic.txt`), and a similar menu is displayed:

1. Encoding
2. Feasibility checking
3. Show the Table
4. Exemplification
5. Omni-optimization
6. Back to previous menu

Figure 6 shows sample console interactions for qualitative logic.

```

Choose an option:
1. Encoding
2. Feasibility checking
3. Show the Table
4. Exemplification
5. Omni-optimization
6. Back to previous menu
Enter your choice: 4
Exemplification:
Two randomly selected feasible objects are o5 and o3,
and they are equivalent in preference.

Choose an option:
1. Encoding
2. Feasibility checking
3. Show the Table
4. Exemplification
5. Omni-optimization
6. Back to previous menu
Enter your choice: 5
All optimal objects: o0, o4

Choose an option:
1. Encoding
2. Feasibility checking
3. Show the Table
4. Exemplification
5. Omni-optimization
6. Back to previous menu
Enter your choice: 6
1. Penalty Logic
2. Qualitative Logic
3. Exit
Choose an option: 2
You picked Qualitative Logic.
Enter the preference file name: TestCase/qualitativechoicelogic.txt

```

Figure 4: Qualitative Logic Menu and Sample Outputs

- **Encoding:** Lists all objects as before.
- **Feasibility Checking:** Displays the count of feasible objects.
- **Exemplification:** Compares two randomly selected objects.
- **Omni-optimization:** Lists all undominated (optimal) objects.

**Example Result for Qualitative Logic:**

Two randomly selected feasible objects are o4 and o11, and they are incomparable in preference.

## 4 Testing Results

### 4.1 Penalty Logic

- **Total feasible objects:** 9
- The table shows individual rule penalties and overall totals.
- Exemplification shows different outcomes (strict preference, equivalence, or incomparability).
- Omni-optimization identifies the objects with the minimal total penalty (e.g., o0, o4).

### 4.2 Qualitative Choice Logic

- **Total feasible objects:** 9
- Exemplification indicates pairwise comparisons with clear outcomes.
- Omni-optimization lists all undominated objects (e.g., o7, o8, o9, o11).

The screenshots (Figures 3 and 6) support the correctness and completeness of the system for both preference logic types.

## 5 Conclusion

The PrefAgent system successfully demonstrates:

1. Encoding of objects from binary attributes.
2. Feasibility checking using hard constraints with PySAT.
3. Preference evaluation using both penalty logic and qualitative choice logic.
4. Clear output in tabular form and through interactive exemplification.
5. Identification of optimal solutions.

Images that match outputs mentioned above are included further below:

```

Choose an option:
1. Encoding
2. Feasibility checking
3. Show the Table
4. Exemplification
5. Omni-optimization
6. Back to previous menu
Enter your choice: 1
Encoding of Objects:
o0 - ice-cream, beer, beef, chips
o1 - ice-cream, beer, beef, fries
o2 - ice-cream, beer, fish, chips
o3 - ice-cream, beer, fish, fries
o4 - ice-cream, wine, beef, chips
o5 - ice-cream, wine, beef, fries
o6 - ice-cream, wine, fish, chips
o7 - ice-cream, wine, fish, fries
o8 - cake, beer, beef, chips
o9 - cake, beer, beef, fries
o10 - cake, beer, fish, chips
o11 - cake, beer, fish, fries
o12 - cake, wine, beef, chips
o13 - cake, wine, beef, fries
o14 - cake, wine, fish, chips
o15 - cake, wine, fish, fries

```

```

Choose an option:
1. Encoding
2. Feasibility checking
3. Show the Table
4. Exemplification
5. Omni-optimization
6. Back to previous menu
Enter your choice: 2
Feasible Objects (9):

```

Figure 5: Qualitative Logic Menu and Sample Outputs

```
Choose an option:
1. Encoding
2. Feasibility checking
3. Show the Table
4. Exemplification
5. Omni-optimization
6. Back to previous menu
Enter your choice: 4
Two randomly selected feasible objects are o4 and o11,
and they are incomparable in preference.

Choose an option:
1. Encoding
2. Feasibility checking
3. Show the Table
4. Exemplification
5. Omni-optimization
6. Back to previous menu
Enter your choice: 5
All optimal objects: o7, o8, o9, o11

Choose an option:
1. Encoding
2. Feasibility checking
3. Show the Table
4. Exemplification
5. Omni-optimization
6. Back to previous menu
Enter your choice: 6
1. Penalty Logic
2. Qualitative Logic
3. Exit
Choose an option: 3
Exiting the PrefAgent system. Goodbye!
```

Figure 6: Qualitative Logic Menu and Sample Outputs

**End of Report**